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PALEOMAGNETISM OF A PALEOCENE PLUTON ON JAMAICA

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A Paleocene granodiorite pluton on Jamaica has been subject to extensive weathering caused by the tropical marine environment of the island. The natural remanence of 29 samples obtained from relatively fresh rock in two localities was found to consist of two components with overlapping coercivity ranges. Alternating field treatment proved ineffective for removing the secondary component without destroying the primary one. Thermal demagnetization of samples from the two localities was more effective and yielded paleomagnetic poles at 14.7° N, 11.6° W and 58.9° N, 15.9° E respectively. These pole positions are different from those available from contemporaneous North American rocks and from poles derived from Jamaican Cretaceous and Upper Miocene rocks. Mineralogical studies showed that the granodiorite has undergone an extensive maghemitization superposed on earlier class 2 deuteric oxidation and related to the weathering process. Some of the titanomaghemite has, however, been converted to titanohematite. Hence although the secondary remanence carried by the former was removable by thermal treatment at 500° C, its part carried by the latter could not be removed without simultaneously destroying the primary remanence carried by the residual titanomagnetite. The observed paleopole positions do not, therefore, represent the true Paleocene geomagnetic field, but suggest that the direction of magnetization of the pluton has been approximately equatorial and was probably acquired in a reversed geomagnetic field. This could be interpreted as having been caused by the behavior of the geomagnetic field during a polarity transition, but a more favorable interpretation appears to be a large anticlockwise tectonic rotation of the islands since the Paleocene.

1. Introduction

Jamaica is one of the Greater Antilles islands which constitute the northern part of the Caribbean arc. The geotectonic history of Jamaica is closely related to other features of the Caribbean. Although focal mechanism studies of Caribbean earthquakes [1] have brought significant evidence about the plate tectonic character of the Caribbean and its present-day easterly drift relative to the Atlantic–North America plate, little is known about the origin of the Caribbean "plate" and its movements in the past.

It is probable that significant paleomagnetic data from Jamaica could resolve some of the ambiguities of the paleotectonics of the Caribbean and provide information on its origin. Recent paleomagnetic studies of Jamaica [2–5] conducted with this aim in mind have been restricted to the Cretaceous and Miocene rocks. The absence of any intermediate time marker between these widely separated geological periods, has given rise to ambiguities in the interpretation of the results. The data can be interpreted in terms of a translational motion of Jamaica from the southwest and/or an anticlockwise rotation since at least the Upper Miocene. Some of the data can also be interpreted in terms of a specific behavior of the geomagnetic field [2]. There exists thus a need for more paleomagnetic information, in particular from rocks of an intermediate age. Unfortunately, the Eocene rocks of Jamaica occur in a tectonically disturbed region of the island and do not give consistent results [6]. The Paleocene pluton of the Above Rocks district, situated to the west of the Eocene outcrops has, on the other hand, undergone little tectonic deformation and could be used as an additional paleomagnetic time marker. This pluton was selected as the subject of our investigation.

2. Geology

The pluton consists mainly of coarse- to medium-grained granodiorite which in places grades into tonalite. It has been dated radiometrically at 65 ± 5 m.y. [7] and is the only early Tertiary intrusion on the island. It crops out over an area of over 55 km^2 and was intruded into the country rock of the Cretaceous Yellow Limestone formation [8]. Unfortunately, in view of the low latitude of the island and its tropical marine environment, the pluton has been subjected to extensive weathering. Only a few exposures exist which are suitable for paleomagnetic sampling. Even these relatively fresh exposures show signs of alteration of the mineral content. The weathering has been superposed on earlier hydrothermal alteration of the rock which has also undergone some deuteric oxidation at the time of its formation. The low-temperature oxidation process, associated with the weathering, has produced profound changes in the magnetic mineralogy of the rock and would thus be expected to have modified its original remanent magnetization.

3. Sampling

Despite the extensive weathering of the pluton a few relatively fresh exposures were found in two localities on the banks of a stream flowing across the outcrop. Paleomagnetic sampling was conducted at these two localities referred to as A and B. A total of 32 oriented samples was obtained, 25 from locality B and seven from A, but due to friability of the rock some samples were destroyed during processing in the laboratory and the total number was reduced to 29, seven from locality A and 22 from B.

4. Experimental results

Measurements of intensity of magnetization were conducted using a Princeton Applied Research Inc. (PAR) Model SM-1 spinner magnetometer. The maximum error of measurement was ± 2 × 10^{-6} emu cm^{-3} in magnitude and about ± 1° in direction. Alternating field (AF) demagnetization was conducted using apparatus capable of developing peak fields of 5000 Oe, while thermal demagnetization was carried out using equipment previously described by Vincenz et al. [9].

The results of observations are summarized in Fig. 1 and Table 1. The data show that the mean direction of natural remanent magnetization (NRM) of specimens from locality A, makes approximately 45° with the present-day geomagnetic field, while the directions of samples from locality B are grouped close to the present-day field direction. The rock clearly carries a secondary remanence partially masking the original remanence. As the table shows, the rock is quite magnetic, implying the presence of a relatively high fraction of magnetic material. Both alternating field (AF) and thermal demagnetization techniques were applied to remove the secondary magnetization as well as for the purpose of studying the magnetic properties of the rock, which were additionally investigated by measurements of Curie temperatures.

AF demagnetization of pilot specimens revealed that the NRM was reasonably stable. Only 25% of its virgin magnitude was destroyed at peak fields of 44 Oe and the specimens still retained about 25–30% at 350 Oe. At 1400 Oe the total magnetization was reduced to one-tenth of its virgin value without a

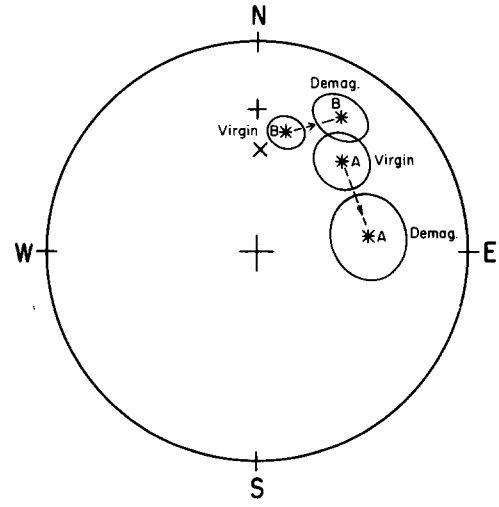


Fig. 1. Changes in the mean NRM directions of locality A and B samples on thermal demagnetization.

TABLE 1
Paleomagnetic results

	Local-	N	D	I	J_n	$\alpha 95$	k	λ'	ϕ'	δ_p	δ_m
Virgin	A	7	43.3	41.0	923.7 ± 201.7	10.6	20.9	44.6°N	3.7°W	8.2°	12.8°
	B	22	13.3	41.6	241.5 ± 17.4	6.6	21.1	76.3°N	14.8°W	5.1°	8.1°
Demagnetized	A	6	82.4	45.9		16.1	20.4	14.7°N	11.6°W	13.6°	20.6°
	B	17	32.1	26.3		9.1	15.9	58.9°N	15.9°E	5.5°	9.8°

Note: N is the number of specimens; D, the declination, I, the inclination, and $\alpha 95$ the radius of circle of confidence, all given in degrees of arc; J_n is the intensity of magnetization in 10^{-6} emu cm^{-3}; k, the precision parameter; λ' is latitude of the paleopole, ϕ' its longitude and δ_p and δ_m the semi-angles of the oval of confidence.

significant change in direction. The directional changes were typified by movement away from the present-day field direction at peak fields up to 350 Oe but higher peak fields caused the directions to return close to their virgin values. It is evident that the secondary components of remanence possessed a stability comparable to that of the primary components and as a result it was not possible to remove the former without simultaneously destroying the latter. The AF technique was, thus, ineffective as means for removing the secondary magnetization and thermal demagnitization was applied. This technique proved more effective for specimens of both localities.

The temperatures to which the samples were heated varied between 420° to 515°C, the optimum temperature being generally 500°C. Thermal "cleaning" caused the direction of magnetization to move away from the present-day field direction and this effect was more pronounced in specimens from locality A than in those from locality B.

Table 1 and Fig. 1 summarize the results of thermal treatment, giving the optimum mean direction obtained by application of the minimum scatter principle. The scatter increased, however, on demagnetization, as shown by the values of α_{95} in both groups of specimens. The final directions were no doubt affected by the uncompensated residual field (about 50 gammas) in the thermal demagnetizer, but its effect was too small to account for the observed scatter which was more likely the result of the overlapping of the blocking temperature ranges of the primary and secondary components of remanence. The secondary component would be expected to possess a part with high blocking temperatures so that only its partial removal was possible without destroying the primary component. After removing the part with the lower blocking temperatures, the presence of the varying proportions of the part with the high blocking temperatures would lead to the observed scatter. This mechanism is justified by the results of petrographic studies given below.

The table shows that the number of specimens in both groups had to be further reduced for statistical computation after thermal "cleaning". This became necessary because some of the "clean" samples became too weakly magnetized to allow reproducible measurements while others disintegrated on heating.

Identification of the magnetic constituents in the rock was conducted by Curie point determinations and microscopic studies of polished sections. The observed Curie points were 560°, 562°, 565°C and 595°, 640° and 675°C. The former three were most probably associated with titanomagnetite while the latter three clearly represented minerals of the ilmenite-hematite series. Microscopic studies revealed an abundance of magnetic opaques which ranged in size from 250μ to 1000μ. Additionally, small grains were present, some as small as 10–20μ. While the sections made from locality B specimens revealed an almost exclusive predominance of large size grains, sections from locality A showed a significant percentage of smaller grains. All grains consisted of titanomagnetite or magnetite partially oxidized to titanomaghemite or maghemite and titanohematite and/or hematite. In each case there was clear evidence of deuteric oxidation with typically class 2 oxidation [10] index. Additionally, the opaques showed presence of superposed light hydrothermal alteration and of pronounced low-temperature oxidation which was indicated by the abundance of maghemite partially oxidized to hematite.

The hydrothermal alteration characteristics of the rock were confirmed also by thin-section studies using transmitted light. An abundance of decomposed or partly decomposed ferromagnesian minerals was observed. Turner [11] has described a natural process whereby the decomposition products of ferromagnesian minerals eventually break down into magnetite when heated in an oxidizing atmosphere. It would appear that in the present case a similar effect has taken place in the laboratory and that heat treatment has "generated" some magnetite from the decomposition products of the ferromagnesian minerals already present in the rocks. This was suggested by the results of experiments summarized in Fig. 2 which shows thermomagnetic curves and curves describing imposition of isothermal remanence. Magnetic material extracted from the granodiorite underwent a decrease in saturation magnetization after cooling from the Curie temperature (Fig. 2a). This was evidently caused by oxidation of magnetite to hematite. On the other hand, whole rock chips subjected to similar treatment revealed an increase of saturation magnetization (Fig. 2b), indicating that fresh magnetite has been produced. Measurement of isothermal remanent magnetization (IRM) in fields up to 10 kOe confirmed

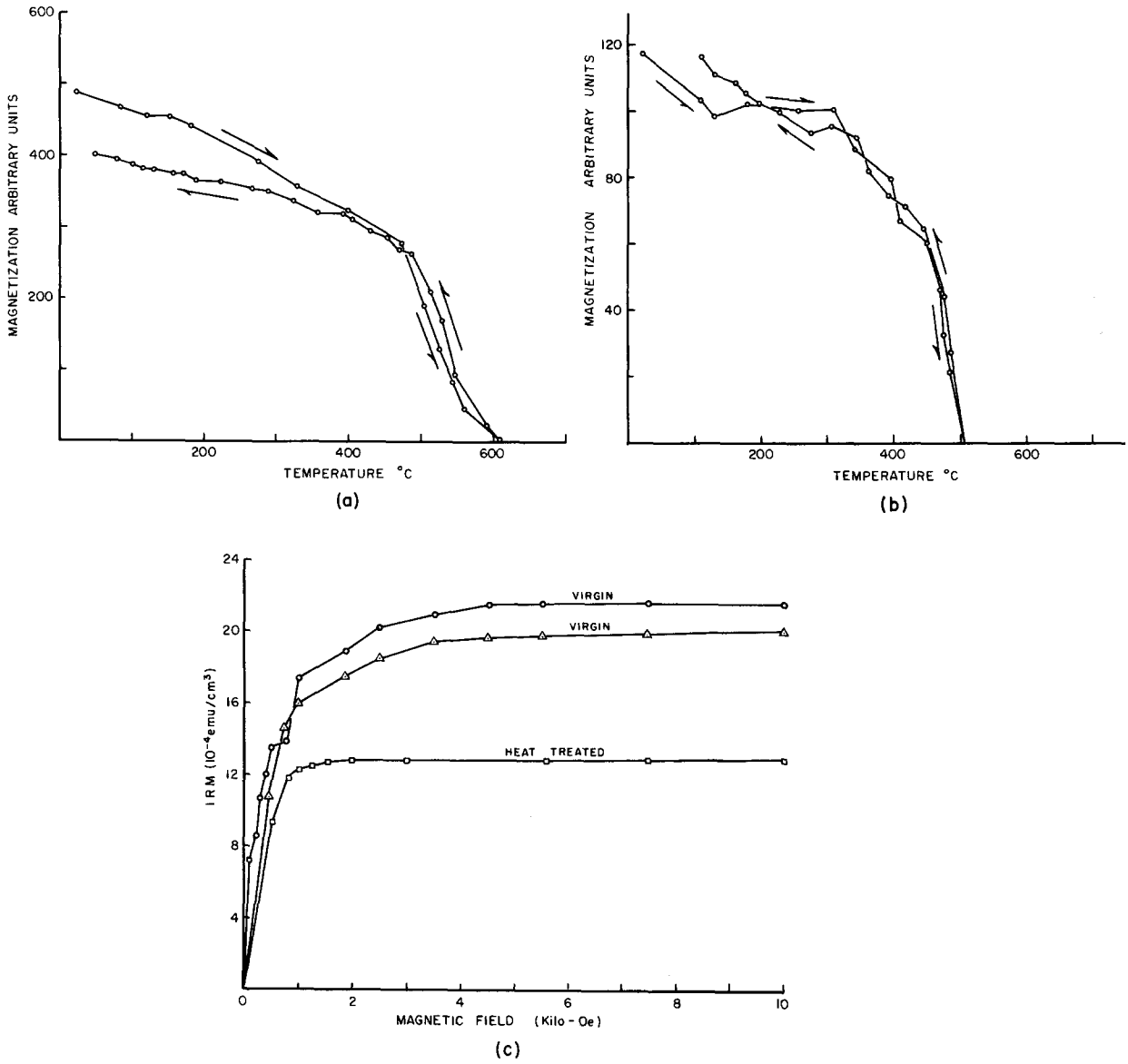


Fig. 2. (a) Thermomagnetic curve for magnetic minerals extracted from virgin rock (Curie temperatures: 560°, 562° and 565°C; (b) Thermomagnetic curve obtained using whole-rock chips; (c) Mode of acquisition of IRM by virgin and baked specimens.

the above deductions. Rock samples previously heated in the laboratory required much smaller fields for saturation than the virgin samples. This suggested that heating resulted in an increase of magnetite content, reducing the effect of hematite originally present in the rock (Fig. 2c). It is concluded that thermal treatment of acid igneous rocks can result in a breakdown of the decomposition products of ferromagnesian minerals yielding magnetite or titanomagnetite as the final product.

5. Discussion of the results

As shown in Table 1, the paleomagnetic pole computed from thermally "cleaned" samples from locality A is situated close to the equator, but the pole obtained from locality B is at a higher latitude. Fig. 3

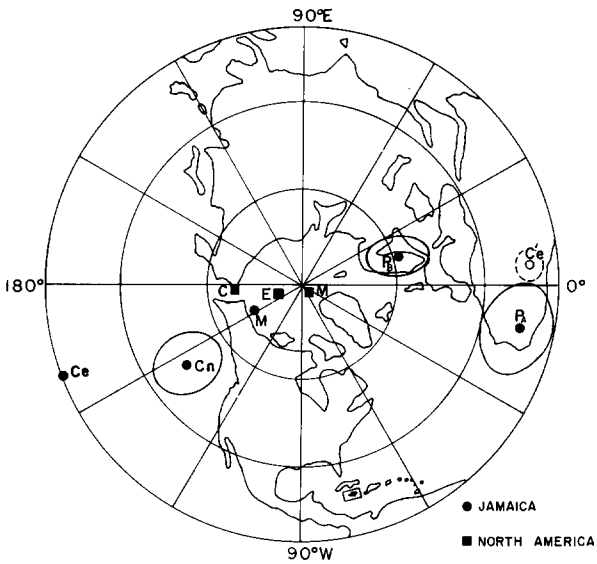


Fig. 3. Paleopole positions of Jamaican and North American rocks. C = Cretaceous; E = Eocene; M = Miocene (the latter after Hicken et al. [12]). Ce and C'e = Jamaican Cretaceous equatorial (2 and 3 sites respectively); Cn = Jamaican Cretaceous "normal" (11 sites) all three poles after Guja and Vincenz [3]; P_A = Jamaican Paleocene locality A; P_B = Jamaican Paleocene locality B (see Table 1 for the values of δ_p and δ_m).

shows the positions of these poles (P_A and P_B respectively) as well as the poles obtained by other authors [3, 5] from Cretaceous rocks on Jamaica. The positions of North American poles based on evidence available to date [12] are also indicated. It is evident that the Paleocene pole positions are different from most of those obtained previously for the Jamaican Cretaceous. Contrary to expectations they do not provide a time marker between the Cretaceous (Cn) and Miocene (M) paleopoles [3]. They differ, however, from the pole positions obtained from the early Tertiary North American rocks and from the present-day geomagnetic field direction. As illustrated in Fig. 3 the paleopole for locality B lies closer to the present-day geomagnetic pole than that for locality A, but both lie on an arc of a circle with Jamaica as a center, which also passes through the North American pole C. This suggests that the secondary magnetization of specimens from locality A was more easily removable than that of specimens from locality B. This suggestion appears justified in terms of the magnetic mineralogy as revealed by the Curie point determination and microscopic studies of polished sections. The studies indicate that the primary remanence of the pluton is carried by class 2 titanomagnetite which has undergone extensive low-temperature oxidation to titanomaghemite and titanohematite. This chemical change resulted in the imposition of secondary magnetization in the form of a chemical remanent magnetization (CRM) as well as viscous remanence (VRM), both acquired recently in a field with a direction close to the present-day field and carried by both titanomaghemite and titanohematite. As pointed out previously, petrographic studies showed that the rock at locality A contained smaller size, less oxidized, titanomagnetite grains than the rock at locality B, indicating that the primary magnetization of the former might be expected to be more stable than that of the latter.

It would appear that the stronger, but thermally unstable secondary magnetization (a CRM and a VRM) carried by titanomaghemite and the weaker, but stable, secondary CRM of the titanohematite compete with the original TRM carried by the residual titanomagnetite. This explains the overlapping coercivity and blocking temperature ranges. The stability of the CRM carried by the titanohematite is clearly comparable with or better than that of the original magnetization. This is reflected in the behavior of the NRM on AF treatment and its ineffectiveness in removing the secondary magnetization. Thermal demagnetization was, on the other hand, more effective because it removed the relatively unstable remanence carried by the titanomaghemite, making the original TRM more apparent. Nevertheless, thermal demagnetization could not remove the CRM carried by titanohematite because its blocking temperatures were comparable with or higher than those of the primary titanomagnetite. Thus on reaching the blocking temperature of the titanomagnetite, the original TRM was destroyed and the contribution of titanohematite became more apparent. Thermal demagnetization demonstrated, therefore, that there is little chance of selectively removing the CRM without almost completely obliterating the original TRM carried by the residual titanomagnetite. Under these circumstances, even in the case of locality A specimens, which carried more stable remanence, the memory of the original direction of magnetization carried by primary titanomagnetite could not be recovered completely. It is concluded that the

paleopoles P_A and P_B do not represent the true Paleocene geomagnetic field for Jamaica, but imply that relative to the present-day position of the island its direction must have been widely different than its present-day direction.

It may be significant, however, that the pole position P_A is close to the equator. The equatorial paleomagnetic pole obtained from three Upper Cretaceous dikes [3] C′e and a similar one reported by Watkins and Cambray [2], although significantly removed from P_A are close enough to suggest that the primary NRM of rock at site A may have had a similar direction. The movement toward the equator of the NRM direction on thermal demagnetization strongly supports this conjecture. It seems thus probable that the Above Rocks granodiorite is magnetized in an easterly equatorial direction like the three Cretaceous rock units represented by pole C′e but unlike twelve other ones represented by pole Cn [4]. It is thus possible that with more favorable magnetic mineralogy, thermal "cleaning" might have yielded a paleomagnetic pole similar to C′e which can be looked upon as a reversed equatorial pole, if pole Ce is considered normal equatorial. If this were so there are two possible interpretations of the data: either the equatorial direction of magnetization reflects the behavior of geomagnetic field during a polarity transition, an effect suggested by previous work [2, 13, 14], or the island has undergone an anticlockwise rotation of about 90° since the Paleocene, and this would yield an easterly, i.e., reversed paleopole position. (The normal equatorial pole Ce [3] is unfortunately based on only two rock units.) If pole Ce is disregarded there is, of course, the third possible interpretation and that is a clockwise rotation of the island without a reversal.

There are some reasons for believing [14] that many low-latitude paleopoles reflect a specific behavior of geomagnetic field during a polarity transition, but on statistical grounds this is not likely to be correct [15]. The frequent occurrence of equatorial directions of magnetization in Jamaican rocks could not thus be wholly accounted for by polarity changes of geomagnetic field. We favor, therefore, at this time the second interpretation of a large tectonic rotation of the island even though a full removal of secondary remanence was not possible. This conclusion in no way contradicts the possibility of a simultaneous translation of the island from the southwest, suggested by other data from Cretaceous rocks [3–5]. It should, however, be emphasized that caution in tectonic interpretation of paleomagnetic data must be exercised at all times. In the present case there is no assurance that the data represent unambiguously only two geomagnetic field directions and they could very well represent a combination of multiple effects.

It is clear that more data are needed for strengthening the argument in favor of the rotation hypothesis. Recent results from Guajira peninsula in Colombia [16] suggest large-scale rotations on southern margins of the Caribbean. More data are needed from the northern margin of the area and these will have to be sought in the study of rocks of contemporaneous ages on other Caribbean islands. Some preliminary paleomagnetic results reported from Hispaniola [17] do, in fact, suggest large-scale rotations of the Greater Antilles in addition to a translational motion from the southwest.

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